# MINI PROJECT DOCUMENTATION

# AI SOLAR PANEL PERFORMANCE ANALYZER

### A Web-Based Solar Energy Performance Analysis System Using Base44 AI

**Project Type:** B.Tech EEE Mini Project

**Branch:** Electrical and Electronics Engineering (EEE)

## Submitted By

Name: _____

Roll Number: _____

College: _____

Academic Year: 2026–2027

**Live Application**
https://literate-solar-sense-pulse.base44.app

# . CERTIFICATE

This is to certify that _____, a student of Bachelor of Technology in Electrical and Electronics Engineering, has successfully completed the mini project entitled **“AI Solar Panel Performance Analyzer”** as part of the academic requirements. The project is developed as a web-based application using Base44 AI and focuses on solar panel performance analysis, energy estimation, efficiency monitoring, savings analysis, and AI-based recommendations.

| Project Guide | Head of Department |
| --- | --- |
| _____ | _____ |

Date: _____

# . DECLARATION

I hereby declare that the mini project entitled **“AI Solar Panel Performance Analyzer”** is my work carried out as part of my B.Tech Electrical and Electronics Engineering mini project. The web application was developed using Base44 AI, an AI-assisted application development platform. The project is intended for educational and demonstration purposes.

Student Name: _____

Signature: _____

Date: _____

# . ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide and the faculty members of the Electrical and Electronics Engineering department for their guidance and support throughout this mini project. I also thank my college for providing the opportunity to develop this project. I am grateful for Base44 AI, which helped in designing and developing the web-based application using AI-assisted development. Finally, I thank everyone who supported me during the development and testing of this project.

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# 1. INTRODUCTION

## Overview

Solar photovoltaic technology is widely used for generating clean electrical energy. The efficiency and energy output of a solar panel system depend on environmental and electrical conditions. Users may not always know whether their solar system is operating efficiently or why its output has decreased. The AI Solar Panel Performance Analyzer provides a simple web-based solution for analyzing solar panel performance using user-provided information.

## Engineering Areas

The project combines concepts from Electrical Engineering, Renewable Energy, Solar Photovoltaic Systems, Web Applications, Data Analysis, and Artificial Intelligence.

# 2. PROBLEM STATEMENT

Solar panel performance can decrease because of dust accumulation, high temperature, low sunlight, shading, panel aging, and electrical losses. Identifying these issues manually can require technical knowledge. Therefore, there is a need for a simple web application that can analyze solar panel parameters and provide understandable performance information and recommendations.

# 3. OBJECTIVES

• Develop a web-based solar panel performance analyzer.

• Analyze electrical and environmental parameters.

• Estimate solar energy generation and system efficiency.

• Calculate energy losses and provide a solar performance score.

• Identify possible causes of poor performance.

• Provide AI-based recommendations and estimated electricity savings.

• Present results using a simple dashboard.

# 4. EXISTING SYSTEM

Conventional solar monitoring systems may require specialized monitoring equipment or technical knowledge. Some systems provide only raw measurements such as voltage, current, and energy generation.

## Limitations:

• Difficult for beginners to understand.

• May require specialized equipment.

• Limited explanation of performance problems.

• Raw data may not provide useful recommendations.

• Advanced monitoring systems can be expensive.

# 5. PROPOSED SYSTEM

The proposed system is a web-based AI Solar Panel Performance Analyzer. The user enters relevant solar system parameters. The application processes the information and provides an estimated performance analysis.

User → Solar Data Input → Data Processing → Performance Analysis → AI-Based Diagnosis → Recommendations → Dashboard & Results

# 6. SCOPE OF THE PROJECT

The project can be used for educational purposes, B.Tech EEE projects, solar energy demonstrations, basic solar performance analysis, understanding renewable energy systems, estimating energy generation, and estimating electricity savings. The current system is primarily intended as an educational and analytical tool.

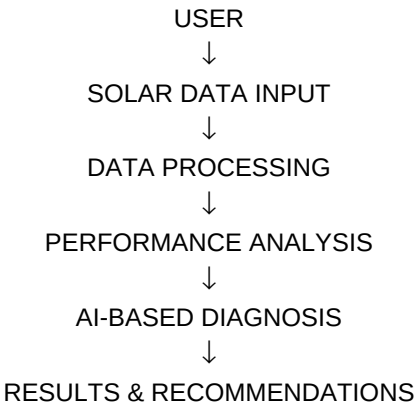
# 7. SYSTEM REQUIREMENTS

| Hardware Requirements | Software Requirements |
|---|---|
| Computer or smartphone | Google Chrome / Microsoft Edge / Firefox |
| Internet connection | Base44 AI platform |
| Minimum 4 GB RAM recommended | GitHub for documentation and sharing |

# 8. TECHNOLOGIES USED

**Development Platform:** Base44 AI
**Application:** Web-based responsive interface
**Domain:** Solar photovoltaic systems, renewable energy, electrical engineering, energy calculations, and performance analysis.
**Development Approach:** AI-assisted application development.

# 9. SYSTEM ARCHITECTURE




The architecture begins with user-entered solar, environmental, and electrical data. The application processes the data, performs calculations, generates an estimated performance assessment, and

presents recommendations and visual results.

# 10. WORKING PRINCIPLE

**Step 1:** Data Input – The user enters panel capacity, efficiency, sunlight hours, temperature, dust level, voltage, current, daily energy generation, and related values.

**Step 2:** Data Processing – The application processes the entered values.

**Step 3:** Energy Analysis – The application estimates energy generation and system performance.

**Step 4:** Performance Evaluation – The system generates an estimated performance score and status.

**Step 5:** AI-Based Diagnosis – The application identifies possible factors affecting performance.

**Step 6:** Recommendations – The system provides suggestions based on the entered data.

# 11. SYSTEM MODULES

## 11.1 Home Module

Introduces the application and provides access to solar analysis features.

## 11.2 Solar Analysis Module

Collects solar panel and environmental information from the user.

## 11.3 Performance Analysis Module

Generates performance score, energy generation, efficiency, energy losses, and system status.

## 11.4 AI Recommendation Module

Provides possible reasons for reduced performance and suggested actions.

## 11.5 Energy Prediction Module

Provides estimated future energy generation.

## 11.6 Savings Analysis Module

Estimates monthly savings, annual savings, and payback period.

## 11.7 Environmental Impact Module

Provides an estimate of environmental benefits such as potential CO■ reduction.

# 12. MATHEMATICAL CALCULATIONS

**Energy Generation:** Energy (kWh) = Power (kW) × Sunlight Hours

**Monthly Energy:** Monthly Energy = Daily Energy × 30

**Annual Energy:** Annual Energy = Daily Energy × 365

**Estimated Savings:** Savings = Energy Generated × Electricity Tariff

These are simplified estimates. Actual solar production depends on system characteristics, irradiance, temperature, shading, losses, and operating conditions.

# 13. AI-BASED ANALYSIS

The application uses AI-assisted logic to provide understandable recommendations. For example, a high dust level combined with reduced energy generation can result in a dust-related performance warning and a recommendation to clean the panel surface. High temperature can similarly result in a temperature-related warning. The application does not physically inspect the solar panel; recommendations are based on user-entered information.

# 14. USER INTERFACE

The application provides a simple and responsive interface containing a home page, solar analysis form, performance dashboard, results, AI recommendations, energy prediction, savings analysis, and environmental impact information.

# 15. TESTING

| Test Case | Input Condition | Expected Result |
|---|---|---|
| 1. Normal | Good sunlight, low dust, moderate temperature | Good/Excellent performance |
| 2. High Dust | Good sunlight, high dust | Dust-related warning and cleaning recommendation |
| 3. High Temperature | High temperature, good sunlight | Temperature-related performance warning |

# 16. ADVANTAGES

• Simple and user-friendly.

• Web-based and accessible through a browser.

• Useful for students and beginners.

• Provides quick performance analysis and visual results.

• Combines EEE, renewable energy, and AI concepts.

• Can be expanded with IoT and machine learning.

# 17. LIMITATIONS

• Results depend on user-provided information.

• The application does not directly measure solar panel parameters.

• Real-time sensor integration is not currently included.

• Weather conditions can change actual generation.

• Predictions are estimates.

• AI recommendations should not replace professional inspection.

• Accuracy can be improved using real-world historical datasets.

# 18. FUTURE SCOPE

• **IoT Sensors:** Integrate voltage, current, temperature, irradiance, and energy sensors.

• **Weather API:** Use live weather information to improve generation prediction.

• **Machine Learning:** Train models using historical solar-generation data.

• **Real-Time Monitoring:** Add a live dashboard for actual solar generation.

• **Mobile Application:** Develop an Android application.

• **Predictive Maintenance:** Predict possible faults before major performance degradation.

# 19. RESULTS

The developed application provides an easy method for analyzing solar panel performance. It can provide a solar performance score, estimated energy generation, efficiency information, possible performance issues, AI-based recommendations, estimated savings, and environmental impact information. The project demonstrates the practical application of AI-assisted web development in Electrical and Electronics Engineering.

# 20. CONCLUSION

The AI Solar Panel Performance Analyzer is a web-based application developed to provide a simple and understandable analysis of solar panel performance. The project combines renewable energy concepts with AI-assisted web application development. The application helps users understand estimated solar generation, efficiency, possible performance issues, savings, and environmental benefits. The system can be enhanced in the future through IoT sensors, weather APIs, real-time monitoring, machine learning, and predictive maintenance.

# 21. REFERENCES


• NPTEL resources related to Solar Photovoltaic Systems and Renewable Energy.

• Fundamentals of Solar Photovoltaic (PV) systems.

• Electrical Engineering fundamentals related to power and energy.

• Base44 AI – AI-assisted web application development platform.

• Renewable energy and solar energy technical resources.


# 22. LIVE APPLICATION

**Project:** AI Solar Panel Performance Analyzer
**Development Platform:** Base44 AI
**Domain:** Electrical and Electronics Engineering – Renewable Energy
**Live Demo:** https://literate-solar-sense-pulse.base44.app

# 23. PROJECT INFORMATION

| Item | Details |
|---|---|
| Project Title | AI Solar Panel Performance Analyzer |
| Branch | Electrical and Electronics Engineering |
| Project Type | B.Tech Mini Project |
| Application Type | Web Application |
| Development Platform | Base44 AI |
| Domain | Solar Energy / Renewable Energy |
| AI | AI-Assisted Analysis |
| Live Demo | https://literate-solar-sense-pulse.base44.app |